

SHUAIJIE SHE

LLM Post-Training · RL · Reasoning · Self-Supervised

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Focus on constructing **self-supervised** learning signals and exploring **general post-training strategies** across tasks, striving toward more **general** and **scalable** LLMs.

🎓 Education

Nanjing University · **Ph.D.** in Computer Science & Technology 2022.09 – Expected 2027.06

First/co-first author of 6 papers at ICLR, ACL, EMNLP, NAACL, and AAAI

Outstanding Graduate Student · **National Scholarship**

Nanjing University · **B.S.** in Computer Science & Technology 2018.09 – 2022.06

📁 Internship Experience: Seed LLM Post-Training

Seed ALM post-training (launched in Doubao/Lark) · Seed-X (200k+ downloads, launched in TikTok)

ByteDance · Seed

2024.04 – Present LLM / Post-Training / LLM

Seed ALM Post-Training

Post-Training · Launched

- **Metrics & Results:** In just two months, human-eval metrics improved substantially (bad-case rate down **9–32** pp) and supported languages grew from 8 to 22; the model is now **launched in Doubao** and **Feishu (Lark)**.
- **Responsibilities:** Owned the **full post-training pipeline**, covering data, training, automatic evaluation, inference engineering, and final quantization/packaging for release:
 - Built a multi-stage **data processing and synthesis pipeline**, curating **billions of SFT examples**;
 - **Built an RL post-training workflow**, synthesizing preference data;
 - **Built a multi-dimensional automatic evaluation system**, developing new metrics including timbre stability;
 - Developed and deployed multiple **inference strategies**, driving engineering and pipeline improvements as well as evaluation, quantization, and launch.

Seed-X LLM Post-Training

Post-Training · Long-CoT

- **Metrics & Results:** The model covers **756** translation directions, matching ultra-large models such as Gemini-2.5, Sonnet-3.5, and GPT-4o on both automatic metrics and human evaluation. Hugging Face Trending **#6** with **200k+** downloads.
- **Responsibilities:** Owned the post-training pipeline powering the **TikTok launch** and explored long-CoT post-training, contributing **tens of millions** in annualized revenue; generated synthetic pre-training data for the Seed foundation model; validated DuPO-based RL on Seed-X.

Other Training Work: Independently reproduced R1-style RLVR and delivered an internal talk; submitted a PR to OpenRLHF improving training efficiency.

4Paradigm · LLM Group

2023.04 – 2024.04 LLM / Post-Training

- **Responsibilities:** SFT, RLHF, and reasoning-capability optimization for Chinese and English LLMs; LLM pre-training data processing.
- **Engineering Practice:** Model post-training, deployment, and prompt tuning; submitted models consistently ranked first on the company's internal general-capability leaderboard.

🔗 Recent Research: RL & Reasoning & Self-Supervision

DuPO: Self-Supervised RL via Generalized Duality

ICLR 2026

- **Background & Goal:** The goal is to let models **self-evolve across diverse tasks, math reasoning, long-context, multimodal, agentic** without human annotation.
- **Metrics & Results:** **+6.4** points on average across four math-reasoning benchmarks including AIME24, reaching SOTA; Qwen3-4B surpasses ultra-large models such as R1 and Sonnet-4. As a training-free reranker, **+9.3** points; on translation, **+2.1** COMET on average across 756 directions.
- **Method & Role:** Collected and cleaned million-scale RL data, ran **extensive RL post-training**, and built the evaluation and full pipeline.
 - Ran RL training on models of diverse architectures, and initial capabilities (Qwen3, LLaMA3.1, Seed-X).
 - **RL infrastructure upgrades:** parallelized reward computation and merged dual-task sampling, improving training efficiency.

- Extended to **long-context** (LongDu, ACL 2026); exploring **multimodal** and **agentic** post-training.

MAPO: Self-Supervised RL for Capability Transfer

ACL 2024

- **Background & Goal:** LLM reasoning ability is unevenly distributed; synthetic data for tasks where the model underperforms is prone to noise, which limits generalization.
- **Metrics & Results:** **No external annotation required:** +16.2 / 6.1 / 13.3 pp accuracy on three math-reasoning datasets, with even larger gains on the OOD test set.
- **Method & Role:** Proposed using cross-lingual reasoning alignment as the reward signal and ran RL training; **RL infrastructure upgrades:** identified the performance-degradation risk of sharing the actor/value backbone and refactored the TRL training framework.

R-PRM: Reasoning-Driven, Self-Evolving Reward Model

EMNLP 2025

- **Background & Goal:** Traditional process reward models output step scores directly, with limited interpretability and data efficiency; the goal is a reward model that “thinks before judging,” makes full use of existing data to explore and evolve, and scores each rollout step accurately.
- **Metrics & Results:** Outperforms strong baselines by **13.9 / 8.5 F1** on ProcessBench / PRMBench; matches SOTA PRMs with only **15%** of their data; scales flexibly with inference-time budget, improving F1 from 61.3 to **71.7**.
- **Method & Role:** Built the PRM post-training pipeline: a “supervised cold start → self-evolving RL → inference-time scaling” recipe.

LongDu: Self-Supervised Long-Context Post-Training

ACL 2026

- **Background & Goal:** Long-input, long-output tasks suffer catastrophic information loss/errors, while document-level parallel data and human evaluation are costly; the goal is to obtain stable long-context training signals from monolingual documents alone, **recovering and improving long-context capability via RL**.
- **Metrics & Results:** LMT-60-8B +19 BLEU and Qwen3-30B-A3B +6.5 BLEU; gains generalize beyond training lengths and to unseen target languages. Also proposed an annotation-free long-context evaluation strategy.
- **Method & Role:** Proposed the method, prepared the data, and ran all **RL post-training experiments:** designed penalty mechanisms to ensure RL stability, and proposed the new long-context evaluation suite.

Publications

(Technical Report 2025) Seed-X: Building Strong Multilingual Translation LLM with 7B Parameters

ByteDance Seed Team

(ICLR 2026, First Author) DuPO: Enabling Reliable LLM Self-Verification via Dual Preference Optimization

Shuaijie She, Yu Bao, Yu Lu, Lu Xu, Tao Li, Wenhao Zhu, Jianbing Zhang, Shujian Huang, Shanbo Cheng, Lu Lu, Yuxuan Wang

(ACL 2026, Co-First Author) Improving Long-Context Translation via Self-Supervised Dual Learning

Shanbo Cheng*, Shuaijie She*, Yu Bao, Jianbing Zhang, Jiajun Chen, Shujian Huang

(EMNLP 2025, Co-First Author) R-PRM: Reasoning-Driven Process Reward Modeling

Shuaijie She*, Junxiao Liu*, Yifeng Liu, Jiajun Chen, Xin Huang, Shujian Huang

(ACL 2024, First Author) MAPO: Advancing Multilingual Reasoning through Multilingual Alignment-as-Preference Optimization

Shuaijie She, Wei Zou, Shujian Huang, Wenhao Zhu, Xiang Liu, Xiang Geng, Jiajun Chen

(NAACL 2024, First Author) Exploring the Factual Consistency in Dialogue Comprehension of LLMs

Shuaijie She, Shujian Huang, Xingyun Wang, Yanke Zhou, Jiajun Chen

(AAAI 2023, First Author) CoP: Factual Inconsistency Detection by Controlling the Preference

Shuaijie She, Xiang Geng, Shujian Huang, Jiajun Chen

(ACL 2026) A Data-Efficient Path to Multilingual LLMs: Language Expansion via Post-training PARAM Δ Integration into Upcycled MoE

Hao Zhou, Tianhao Li, Zhijun Wang, Shuaijie She, et al.

(ACL 2024) Question Translation Training for Better Multilingual Reasoning

Wenhao Zhu, Shujian Huang, Fei Yuan, Shuaijie She, Jiajun Chen, Alexandra Birch

(EMNLP 2024) Multilingual Contrastive Decoding via Language-Agnostic Layers Skipping

Wenhao Zhu, Sizhe Liu, Shujian Huang, Shuaijie She, Chris Wendler, Jiajun Chen

(EMNLP 2023) Improved Pseudo Data for Machine Translation Quality Estimation with Constrained Beam Search

Xiang Geng, Yu Zhang, Zhejian Lai, Shuaijie She, et al.